CODECHELLA MADRID 2026

Foundations of Difference-in-Differences

Day 1: The 2×2, Parallel Trends, and Event Studies

Scott Cunningham · Baylor University

CUNEF Auditorium · May 25–28, 2026

Welcome to Codechella Madrid 2026

Our third annual workshop here at CUNEF.

Scott Cunningham · Baylor

Kyle Butts · Arkansas

(at home with his new baby!)

with **Mark Anderson** (Montana State)

Dan Rees (UC3M) · **Agustín Casas** (CUNEF)

What we're here for

Practical causal inference.

The methods.

Using them well.

Our opinions, along the way.

The week

	Day 1 Monday, May 25 CUNEF Auditorium	**Day 2** Tuesday, May 26 CUNEF Auditorium	**Day 3** Wednesday, May 27 CUNEF Auditorium	**Day 4** Thursday, May 28 CUNEF Auditorium
8:30 - 9am	**Registration**			
9 — 10:30am	Core DID	Covariates	Continuous	Staggered
10:30 — 11am	Coffee Break	Coffee Break	Coffee Break	Coffee Break
11am — 1pm	Covariates	Covariates	Continuous	Staggered
1 — 2:30pm	Lunch (by Research Field)	Lunch	Lunch	Lunch
2:30 — 3:30pm	*Dan Rees & Mark Anderson* Hidden Curriculum	Continuous	Staggered	Claude Code
3:30 — 5pm	*Dan Rees & Mark Anderson* Hidden Curriculum	Continuous	*Dan Rees & Mark Anderson* Hidden Curriculum	Claude Code

Today: the basics of DiD

Four moving parts.

Calculation.

Identification.

Regressions.

Falsifications.

After today

Covariates	· Mon PM + all of Tue
Continuous DiD	· Tue PM + Wed AM
Staggered DiD	· Wed PM + Thu AM
Hidden Curriculum	· Mon & Wed afternoons, with Mark & Dan
Claude Code	· Thu PM

Lesson

1. DiD's Empirical Roots

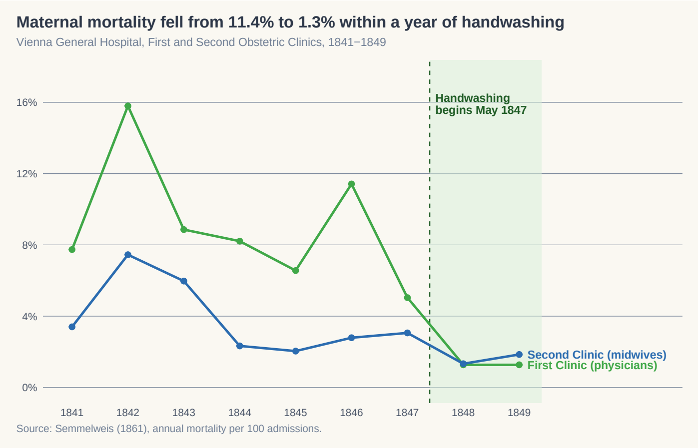
Vienna, 1840s: two clinics, two mortality rates

- Vienna General Hospital, First Obstetric Clinic
- Pregnant women were assigned to the *physicians' wing* or the *midwives' wing* by day of admission — effectively at random
- Physicians' wing: **13–18% mortality** from puerperal fever
- Midwives' wing: **about 3%**
- Women would deliver in the street rather than risk the physicians' wing

His hypothesis, and his DiD

- Semmelweis — an assistant on the wards — noticed the physicians did cadaver dissections before delivering babies. The midwives did not.
- Hypothesis: something on the physicians' hands was causing the fever.
- May 1847: he instituted **chlorinated lime handwashing for physicians only**
- Midwives served as the untreated comparison group — a DiD by hand

The data



Monthly mortality, First Obstetric Clinic. Handwashing began mid-1847.

Princeton, 1970s: Orley Ashenfelter

The Industrial Relations Section was founded 1922 — rigorous empirical labor research, the policy edge of US economics.

Orley Ashenfelter's leadership

- Long-time director. Set the tone: high-quality data, careful design, communicate to bureaucrats.
- Trained a cohort: **David Card**, **Josh Angrist**, **Bob LaLonde**, **Janet Currie**, Pischke, Dinardo, Case, others.
- Hired **Alan Krueger** as a junior colleague.

By the late 1970s, empirical labor was in trouble

- Different methods on the same question disagreed wildly
- The Princeton response: **design over modeling — estimators that mimic an RCT as closely as possible**
- Worker surveys, federal job training, massive panel regressions
- Audience was bureaucrats

> *Their eyes would glaze over whenever you said 'regression.'*

— Orley Ashenfelter

youtu.be/WnB3EJ8K7lg *(3:53)*

LESSON

2. DiD is Two Things: a Calculation and an Assumption

DiD is a research design, not a regression

> *A research design is an identification strategy: it reveals the invisible target parameter hidden inside a calculation.*

1.		2.
A *calculation*	**+**	**An *assumption***
(the 2×2)		(parallel trends)

The individual treatment effect

$$\delta_i \;=\; Y_i(1) - Y_i(0)$$

- One per unit. **Different** across units in general — treatment effects are heterogeneous.
- We only ever see one of $Y_i(1)$ or $Y_i(0)$. The other is the counterfactual.
- So δ_i is never directly observed at the individual level.

The ATT: a causal parameter has three ingredients

$$\text{ATT} = \underbrace{\mathbb{E}\Big[}_{\text{weighted expectations}} \underbrace{Y(1) - Y(0)}_{\text{potential outcomes}} \underbrace{\Big| D = 1, \text{Post}\Big]}_{\text{population}}$$

- **Potential outcomes** — the contrast $Y(1) - Y(0)$
- **Population** — on whom: treated units, post-period
- **Weighted expectations** — how we average across that population ($\mathbb{E}[\cdot]$ defaults to equal weights)

The switching equation: you see one half

$$Y_i \;=\; D_i \cdot Y_i(1) \;+\; (1 - D_i) \cdot Y_i(0)$$

- For each unit: $Y_i(1)$ if treated, $Y_i(0)$ if not.
- The ATT asks about *both* columns. The data hands you one.

The rest of the day: strategies for recovering the ATT from the half you see.

LESSON

3. Five Regressions, One Number

Cheng & Hoekstra (2013): castle doctrine and homicide

The reform. "Castle doctrine" laws expanded the legal right to use lethal force in self-defense, removing the duty to retreat. Many US states passed expansions from **2005–2010**.

- **Panel:** all US states $\times$ years (state-year).
- **Outcome:** log homicide rate, FBI Supplementary Homicide Reports.
- **Treatment:** state passed an expansion of castle doctrine.
- **Comparison:** states that did not pass (never-treated).

For today: a clean 2×2 on **2005–2006**. Treated $=$ states adopting in 2006; control $=$ states that hadn't. *Cheng & Hoekstra (2013), "Does Strengthening Self-Defense Law Deter Crime or Escalate Violence? Evidence from Expansions to Castle Doctrine." J. Human Resources 48(3): 821–854.*

The four cells, by hand on `castle.dta`

```
use castle.dta, clear
keep if year >= 2005 & year <= 2006
gen post = (year == 2006)
gen treat = (effyear == 2006)

summarize l_homicide if treat==1 & post==1 // Y11
summarize l_homicide if treat==1 & post==0 // Y10
summarize l_homicide if treat==0 & post==1 // Y01
summarize l_homicide if treat==0 & post==0 // Y00
```

$$\widehat{\mathsf{DiD}} = (\overline{Y}_{11} - \overline{Y}_{10}) - (\overline{Y}_{01} - \overline{Y}_{00}) \approx \mathbf{0.10}$$

Log homicide rate $\sim 10\%$ *higher in 2006-adopting states relative to non-adopters.*

Regression 1: OLS with treat $\times$ post

```
reg l_homicide post##treat, cluster(sid)
```

The coefficient on `1.post#1.treat` is the DiD.

$$Y = \beta_0 + \beta_1 \text{Post} + \beta_2 \text{Treat} + \beta_3 (\text{Post} \times \text{Treat}) + \varepsilon$$

The saturated specification. $\hat{\beta}_3 \approx 0.10$.

Regression 2: two-way fixed effects

```
xtset sid year
xtreg l_homicide c.treat#c.post i.year, fe vce(cluster sid)
```

State fixed effects absorb time-invariant level differences. Year fixed effects absorb common shocks.

In the 2×2 case, the TWFE coefficient equals the DiD. Same number.

Regression 3: first-difference ΔY_i on D_i

```
keep sid year l_homicide treat
reshape wide l_homicide, i(sid) j(year)
gen dY = l_homicide2006 - l_homicide2005
reg dY treat, vce(cluster sid)
```

$$\Delta Y_i \;=\; \alpha + \beta_{DD}\, D_i + u_i$$

- Collapse the panel: one ΔY per unit.
- Time-differencing wipes out unit fixed effects mechanically.
- Slope on `treat` = the DiD. Same number.

Regression 4: $(Y_{\text{treat}} - Y_{\text{ctrl}})$ on the post dummy

At each time t, take the across-group difference $D_t = \overline{Y}_{1,t} - \overline{Y}_{0,t}$. Then regress on a post indicator:

$$D_t \;=\; \alpha + \beta_{DD}\,\text{Post}_t + v_t$$

```
collapse (mean) l_homicide, by(treat year)
reshape wide l_homicide, i(year) j(treat)
gen Dt = l_homicide1 - l_homicide0
gen post = (year == 2006)
reg Dt post
```

- Regression 3 differences over *time*; Regression 4 differences over *groups* first.
- By Frisch–Waugh–Lovell, partialling out the two-way means runs in either order. Same number.

Regression 5: long differences — the Snow move

Compute one number per group, post minus pre, then subtract:

$$\widehat{\text{DiD}} = \left(\overline{Y}_{11} - \overline{Y}_{10}\right) - \left(\overline{Y}_{01} - \overline{Y}_{00}\right)$$

- No regression. Two long differences. One subtraction between them.
- This is what John Snow did in 1854 with the water-company data.
- Identical to Regressions 1–4. **The arithmetic was the point all along.**

Five regressions, one number

Specification	What it does	$\hat{\beta}_{DD}$
1. OLS with `post##treat`	saturated interaction	≈ 0.10
2. Two-way fixed effects	absorb unit + time means	≈ 0.10
3. ΔY_i on D_i (first difference)	difference over time, then on D	≈ 0.10
4. D_t on Post$_t$ (across-group diff)	difference over groups, then on Post	≈ 0.10
5. Long differences (Snow)	two means, one subtraction	≈ 0.10

Same data, five specifications, identical to machine epsilon.
The 2$\times$2 is the object; the regressions are just five ways to ask for it.

Lesson

4. Decomposing the 2×2

The proof: add a zero, rearrange (five steps)

1. Start with the 2×2:

$$\widehat{\text{DiD}} \;=\; \big[\mathbb{E}[Y \mid 1, \text{Post}] - \mathbb{E}[Y \mid 1, \text{Pre}]\big] \;-\; \big[\mathbb{E}[Y \mid 0, \text{Post}] - \mathbb{E}[Y \mid 0, \text{Pre}]\big]$$

2. Switching equation. Treated-post is $Y(1)$; the other three cells are $Y(0)$:

$$\widehat{\text{DiD}} \;=\; \big[\mathbb{E}[Y(1) \mid 1, \text{Post}] - \mathbb{E}[Y(0) \mid 1, \text{Pre}]\big] \;-\; \big[\mathbb{E}[Y(0) \mid 0, \text{Post}] - \mathbb{E}[Y(0) \mid 0, \text{Pre}]\big]$$

3. Add a zero. Insert $+\,\mathbb{E}[Y(0) \mid 1, \text{Post}] - \mathbb{E}[Y(0) \mid 1, \text{Post}]$:

$$\begin{aligned}\widehat{\text{DiD}} &= \mathbb{E}[Y(1) \mid 1, \text{Post}] \;-\; \mathbb{E}[Y(0) \mid 1, \text{Post}] \\ &\quad + \; \mathbb{E}[Y(0) \mid 1, \text{Post}] - \mathbb{E}[Y(0) \mid 1, \text{Pre}] \;-\; \big[\mathbb{E}[Y(0) \mid 0, \text{Post}] - \mathbb{E}[Y(0) \mid 0, \text{Pre}]\big]\end{aligned}$$

4. Rearrange. First two terms = ATT; remainder = difference in $Y(0)$ trends:

$$\widehat{\text{DiD}} \;=\; \underbrace{\mathbb{E}[Y(1) - Y(0) \mid 1, \text{Post}]}_{\textbf{ATT}} \;+\; \underbrace{\big\{\Delta\mathbb{E}[Y(0) \mid 1] \;-\; \Delta\mathbb{E}[Y(0) \mid 0]\big\}}_{\text{non-parallel-trends bias}}$$

5. Read. If the bias equals zero (*parallel trends*), then $\widehat{\text{DiD}} = \text{ATT}$.

The bias term *is* selection bias — on trends, not levels

The "non-parallel-trends bias" isn't a new concept. It's selection bias, in a different costume.

$$\text{bias} = \underbrace{\Delta\mathbb{E}[Y(0) \mid D{=}1]}_{\text{how treated's } Y(0) \text{ moves}} - \underbrace{\Delta\mathbb{E}[Y(0) \mid D{=}0]}_{\text{how controls' } Y(0) \text{ moves}}$$

- **Cross-section selection bias**: are treated and control different in *levels* of $Y(0)$?
- **DiD's selection bias**: are treated and control different in *trends* of $Y(0)$?
- Different levels are fine. Unit fixed effects soak them up.
- What can't differ is the *rate* at which $Y(0)$ moves across groups.

The trade DiD makes vs. randomization

RCTs: treated and controls identical in everything — levels and trends.
DiD: identical only in *trends* of $Y(0)$. **Cheaper assumption. Weaker identification.**

Case 1: NA holds, never-treated comparison

Switching equation. Treated-post is $Y(1)$; the other three cells are $Y(0)$.

$$\widehat{\text{DiD}} = \big[\mathbb{E}[Y(1)|1,\text{Post}] - \mathbb{E}[Y(0)|1,\text{Pre}]\big] - \big[\mathbb{E}[Y(0)|0,\text{Post}] - \mathbb{E}[Y(0)|0,\text{Pre}]\big]$$

Add the zero. Insert $+\mathbb{E}[Y(0)|1,\text{Post}] - \mathbb{E}[Y(0)|1,\text{Post}]$. Rearrange:

$$\widehat{\text{DiD}} = \underbrace{\mathbb{E}[Y(1) - Y(0)|1,\text{Post}]}_{\textbf{ATT}} + \underbrace{\big\{\Delta\mathbb{E}[Y(0)|1] - \Delta\mathbb{E}[Y(0)|0]\big\}}_{\text{non-PT bias} = 0 \text{ under PT}}$$

Numerical example

ATT $= 10$, PT bias $= 0$. $\widehat{\text{DiD}} = \mathbf{10}$. ✓

Case 2: NA fails — anticipation contaminates Y_{Pre}

Suppose the treated group anticipates and adjusts *before* the policy lands. Their pre-period outcome contains some $Y(1)$ response: $\mathbb{E}[Y|1,\text{Pre}] = \mathbb{E}[Y(0)|1,\text{Pre}] + \text{ATT}_{\text{pre}}$.

Same +0 trick, propagated:

$$\widehat{\text{DiD}} = \underbrace{\text{ATT}_{\text{Post}}}_{\text{what we want}} + \underbrace{\text{non-PT bias}}_{=0} - \underbrace{\text{ATT}_{\text{pre}}}_{\text{anticipation}}$$

Numerical example

$\text{ATT} = 10$, PT bias $= 0$, $\text{ATT}_{\text{pre}} = 3$. $\widehat{\text{DiD}} = 10 - 3 = \mathbf{7}$. **biased down**

If treatment effects are roughly constant, DiD can collapse to zero.

Case 3: the control group also receives treatment in Post

Now the comparison group's Post outcome is also $Y(1)$, not $Y(0)$. Switching equation:

$$\widehat{\text{DiD}} = \Big[\mathbb{E}[Y(1)|1,\text{Post}] - \mathbb{E}[Y(0)|1,\text{Pre}]\Big] - \Big[\mathbb{E}[Y(1)|0,\text{Post}] - \mathbb{E}[Y(0)|0,\text{Pre}]\Big]$$

Add two zeros — one missing counterfactual on each side — and rearrange:

$$\widehat{\text{DiD}} = \underbrace{\mathbb{E}[Y(1)-Y(0)|1,\text{Post}]}_{\textbf{ATT}} + \underbrace{\Big\{\Delta\mathbb{E}[Y(0)|1] - \Delta\mathbb{E}[Y(0)|0]\Big\}}_{\text{non-PT bias} = 0 \text{ under PT}} - \underbrace{\mathbb{E}[Y(1)-Y(0)|0,\text{Post}]}_{\textbf{ATT}_{D=0,\text{Post}}\ (spillover)}$$

Numerical example

ATT $= 10$, PT bias $= 0$, ATT$_{D=0,\text{Post}} = 4$. $\widehat{\text{DiD}} = 10 - 4 = \mathbf{6}$. **biased toward zero**

Three cases, three failures of the punchline. The +0 trick still works — only the zero you add changes.

Summary: four ways the 2×2 decomposes

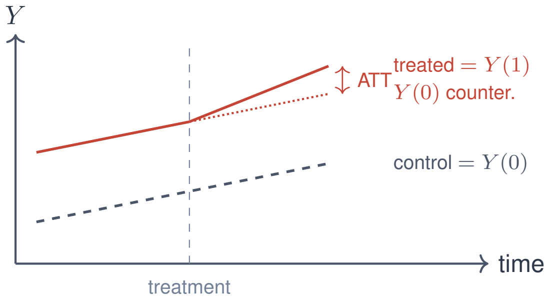
What holds	$\widehat{\text{DiD}} =$
NA, never-treated control	$\text{ATT} + \text{PT-bias}$
PT, no NA, never-treated control	$\text{ATT}_{\text{Post}} - \text{ATT}_{\text{Pre}}$
PT, NA, always-treated control	$\text{ATT} - \Delta\text{ATT}_{D=0}$
PT, NA, SUTVA fails (spillover)	$\text{ATT} - \text{ATT}_{D=1,\text{Post}}$

All four derivable the same way: write down the 2×2, apply the switching equation, add the right series of zeros.

LESSON

5. Card and Krueger: Parallel Trends, Visualized

Parallel trends, drawn



$$\mathbb{E}[Y(0) \mid D{=}1, \text{Post}] - \mathbb{E}[Y(0) \mid D{=}1, \text{Pre}] \;=\; \mathbb{E}[Y(0) \mid D{=}0, \text{Post}] - \mathbb{E}[Y(0) \mid D{=}0, \text{Pre}]$$

A 2×2 of $Y(0)$ terms. ***If it isn't four $Y(0)$s, it isn't the PTA.***

Under PT, the 2×2 hits the ATT

With (i) **No Anticipation**, (ii) **Parallel Trends in $Y(0)$**, and (iii) a clean (never-treated) comparison group:

$$\text{ATT}_{\text{post}} = \underbrace{\mathbb{E}[\Delta Y \mid D = 1]}_{\text{treated change}} - \underbrace{\mathbb{E}[\Delta Y \mid D = 0]}_{\text{control change}}$$

- The 2×2 calculation equals the population's ATT.
- Four averages. Three subtractions. One target parameter.

This is the punchline. Everything that follows is variations on this theme.

New Jersey raises its minimum wage. Pennsylvania doesn't.

- **April 1992**: New Jersey raises its minimum wage from $4.25 to $5.05 — a 19% increase.
- Pennsylvania, just over the river, stays at $4.25.
- Card and Krueger survey 410 fast-food restaurants on both sides of the border, before (Feb 1992) and after (Nov 1992).
- The natural 2×2: **treated $=$ NJ, control $=$ PA, pre/post the policy.**

Card & Krueger, American Economic Review, 1994.

What Card and Krueger *found*

Employment per restaurant	Pre (Feb 1992)	Post (Nov 1992)	Change
NJ (treated)	20.44	21.03	$+0.59$
PA (control)	23.33	21.17	-2.16
DiD (NJ – PA)			**+2.76 jobs/restaurant**

- Employment in NJ fast food *rose* slightly, relative to PA, after the wage hike.
- Standard error puts the effect roughly indistinguishable from zero — but the sign is the opposite of what theory predicted.
- **No measurable disemployment effect.** The textbook had failed a real test.

The reaction was personal

> *No self-respecting economist would claim that increases in the minimum wage increase employment. Such a claim becomes equivalent to a denial that there is even minimal scientific content in economics.*

— James Buchanan (Nobel 1986), *WSJ*, April 1996

> *It cost me a lot of friends. People that I had known for many years became very angry or disappointed. They thought that in publishing our work we were being traitors to the cause of economics as a whole.*

— David Card, interview

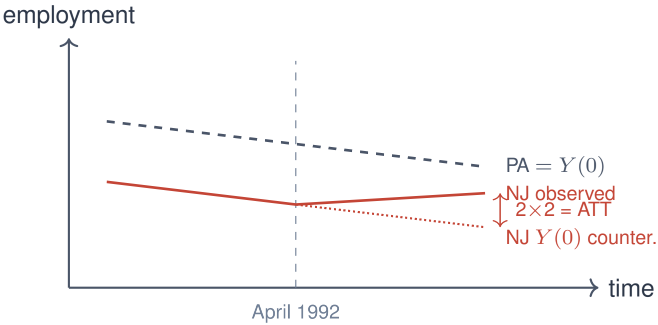
The price of a credible empirical paper that contradicted the consensus. Krueger paid more.

Hear it from them — three short interview clips

- **Orley Ashenfelter on the profession's reaction** (start 31:20)
  `youtu.be/MOtbuRX4eyQ?t=1882`
- **Card on the hostility from peers** (1:07–1:10)
  `youtu.be/1soLdywFb_Q?t=4020`
- **Card on what happened to Alan Krueger** (1:16–1:17)
  `youtu.be/1soLdywFb_Q?t=4556`

The DiD literature did not arrive politely. It was earned through paper-by-paper combat, much of it personal.

Parallel trends visualized: the control *is* the counterfactual



The dotted line is what NJ would have done absent the policy. We never see it.

PT lets PA's slope stand in for it.

Aside: there's also an *across-group* PT we never name

The usual statement (within-group, across-time)

The comparison group's $Y(0)$ trend stands in for what the treated group's $Y(0)$ trend would have been.

The same assumption, stated from the other side (across-group, within-time)

The treated group's $Y(0)$ trend (had they not been treated) would equal the comparison group's $Y(0)$ trend.

- Two phrasings of one assumption: $Y(0)$ trends are common across groups.
- Event-study β_k vs $t = -1$ computes the within-group across-time piece.
- Regression 3 (ΔY on D) and Regression 4 (D_t on Post) are FWL mirrors — same PT from the two directions.

ASIDE — drop in or drop out if behind schedule.

Pre-trends are the gut check — not a proof of PT

- Card–Krueger held up because in the pre-period, NJ and PA fast-food employment moved together.
- If NJ employment had already been diverging before April 1992, the $+2.76$ estimate would have been mostly trend, not policy.
- Flat pre-trends $\rightarrow$ evidence (not proof) the design holds.
- **The 2$\times$2 was credible because of the pre-period parallelism — not because of the regression coefficient.**

More later (Roth, Roth–Sant'Anna): when pre-trends look flat but the test still over-rejects.

Lesson

6. Open the Sheet

Open the Sheet: stability and trends, by hand

Live demo

Compute the four cells, three subtractions, and the by-hand event study — in a spreadsheet.

docs.google.com/spreadsheets/d/1onabpc14JdrGo6NFv0zCWo-nuWDLLV2L1qNogDT9SBw

Four tabs in sequence: `Potential Outcomes` → `2X2` → `Balancing` → `2XT #1`.

LESSON

7. Standard Errors

Three flavors of standard error

- **Vanilla (homoskedastic)**: assumes $\mathrm{Var}(\varepsilon_i \mid X_i) = \sigma^2$ for all i.
- **Robust (Eicker–Huber–White, HC)**: each residual speaks for itself — handles heteroskedasticity.
- **Cluster-robust (CRVE)**: groups residuals by cluster; within-cluster errors correlate freely.

Same point estimate, three different SEs. The SE you pick is a claim about the error structure.

HC1 numerically: each $\hat{e}_i^2$ weighted by leverage

Six observations. Three pairs at $\tilde{x} \in \{-2, 0, +2\}$. Fitted $\hat{\beta}_1 = 1$; residuals shown.

i	cluster g	$\tilde{x}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$\tilde{x}_i^2 \cdot \hat{e}_i^2$
1	1	-2	$+0.5$	0.25	1.00
2	1	-2	$+0.5$	0.25	1.00
3	2	0	-1.0	1.00	0.00
4	2	0	-1.0	1.00	0.00
5	3	$+2$	$+0.5$	0.25	1.00
6	3	$+2$	$+0.5$	0.25	1.00
				meat $=$	**4.00**

With $n = 6$, $k = 2$, and $S_{xx} = \sum_i \tilde{x}_i^2 = (-2)^2+(-2)^2+0+0+2^2+2^2 = 16$:

$$\widehat{\text{Var}}_{\text{HC1}}(\hat{\beta}_1) = \frac{n}{n-k} \cdot \underbrace{\frac{1}{S_{xx}}}_{\text{bread}} \cdot \underbrace{\text{meat}}_{4.00} \cdot \underbrace{\frac{1}{S_{xx}}}_{\text{bread}} = \frac{6}{4} \cdot \frac{4.00}{16 \cdot 16} = \frac{6}{4} \cdot \frac{4.00}{256} = 0.02344$$

$$\Rightarrow \quad \textbf{SE}_{\textbf{HC1}} = \sqrt{0.02344} \approx 0.153$$

Obs 3–4 contribute zero because $\tilde{x} = 0$ kills the leverage weight. ***HC1 treats all 6 obs as independent.***

Why HC-robust barely helps in DiD

The HC correction reweights residuals by $X_i X_i'$. In a DiD with binary $D_i \in \{0, 1\}$ and roughly balanced groups:

- Treated and control units have similar *leverage*: each side carries $\approx$ half the weight in $X'X$.
- Residual variance is roughly symmetric across the two groups (binary D doesn't generate a fan shape in $\hat{e}^2$).
- The weighted sum $\sum_i \hat{e}_i^2 X_i X_i'$ ends up not very different from $\hat{\sigma}^2(X'X)$.

Consequence

$\text{SE}_{\text{HC}} \approx \text{SE}_{\text{vanilla}}$ for typical DiD specifications.
The over-rejection problem is not heteroskedasticity. It's something else.

CRVE numerically: same 6 obs, group scores, then square

Same 6 observations. Now compute $s_g = \sum_{i \in g} \tilde{x}_i \hat{e}_i$ for each cluster, then s_g^2:

i	g	$\tilde{x}_i$	$\hat{e}_i$	$\tilde{x}_i \hat{e}_i$
1	1	-2	$+0.5$	-1
2	1	-2	$+0.5$	-1
3	2	0	-1.0	0
4	2	0	-1.0	0
5	3	$+2$	$+0.5$	$+1$
6	3	$+2$	$+0.5$	$+1$

g	s_g	s_g^2
1	-2	4
2	0	0
3	$+2$	4
meat =		**8**

With $G = 3$, $n = 6$, $k = 2$, and $S_{xx} = \sum_i \tilde{x}_i^2 = 16$, so $S_{xx}^2 = 256$:

$$\widehat{\text{Var}}_{\text{CRVE}}(\hat{\beta}_1) = \frac{G}{G-1} \cdot \frac{n-1}{n-k} \cdot \frac{\text{meat}}{S_{xx}^2} = \frac{3}{2} \cdot \frac{5}{4} \cdot \frac{8}{256} = 0.05859 \Rightarrow \mathbf{SE}_{\mathbf{CRVE}} \approx 0.242$$

***HC1**: 0.153 vs. **CRVE**: 0.242 (**58% larger**). Same data. The cluster cross-terms changed the meat from 4 to 8.*

Bertrand, Duflo & Mullainathan (QJE 2004): the placebo-law test

"How Much Should We Trust Differences-in-Differences Estimates?"

- BDM generate *placebo laws*: assign random treatment dates to states in a real CPS panel.
- True effect by construction: $\beta = 0$. So $\Pr(\text{reject } H_0)$ should equal $\alpha = 5\%$.
- **Vanilla SE**: rejection rate $\approx \mathbf{45\%}$. Wildly over-rejects.
- **Heteroskedasticity-robust (HC1)**: rejection rate *also* $\approx 45\%$. **HC barely moves the needle.**
- **Cluster-robust at state level**: rejection rate $\approx \mathbf{5\%}$. The fix.

Why HC doesn't save you

Binary, balanced D + within-state serial correlation. The het-correction has nothing to do with the actual problem. **Clustering is doing all the work.**

LESSON

8. Population Weighting

Ten people, two counties

Person	$Y(0)$	$Y(1)$	δ_i	**County**
Alan	0	1	1	1
Betty	1	1	0	1
Chad	1	1	0	1
Daniel	0	0	0	1
Edith	0	1	1	1
Frank	0	1	1	1
George	0	0	0	1
Hank	0	1	1	1
Ida	1	0	-1	2
Janet	1	0	-1	2

County 1: eight people. County 2: two people. Compute the average δ for each county, and across all ten. Same answer?

Same data, two weights, two parameters

- **Within counties:** County 1 avg $\delta = \frac{1}{8}(1{+}0{+}0{+}0{+}1{+}1{+}0{+}1) = +\mathbf{0.5}$.
  County 2 avg $\delta = \frac{1}{2}(-1{+}(-1)) = -\mathbf{1.0}$.
- **ATE for the average person**: $\frac{1}{10}(8 \cdot 0.5 + 2 \cdot (-1)) = +\mathbf{0.2}$.
- **ATE for the average county**: $\frac{1}{2}(0.5 + (-1)) = -\mathbf{0.25}$.

Same potential outcomes. ***Different signs.***
Whichever you report depends on what you're trying to answer.

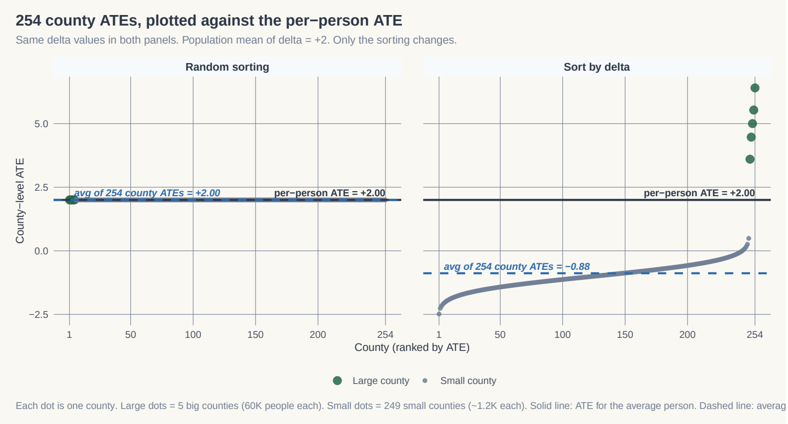
Two parameters, two answers

Scale the example up: **Texas, 254 counties, 30M people.** 5 biggest counties hold 41% of the population. Half the population has $\delta_i \sim \mathcal{N}(+5, 1)$; half has $\delta_i \sim \mathcal{N}(-1, 0.5)$. So $\mathbb{E}[\delta] = +2$. Sort high-δ people into the 5 big counties.

- **Per-person ATE**: $\frac{1}{N} \sum_i \delta_i$ — weight every person by $1/N$.
  15M high-δ balance 15M low-δ. **Answer: $+2$.**
- **Avg of county ATEs**: $\frac{1}{254} \sum_c \bar{\delta}_c$ — weight every county by $1/254$.
  $\frac{5(+5)+249(-1)}{254} =$ **-0.88**. The 249 small counties dominate.

Same data. Two weights. Two parameters. Sometimes two signs.

Sorting on δ can flip the sign of your answer



Random: every county's ATE equals the population's. δ*-sort: per-person line at* $+2$*, avg of county ATEs flips to* -0.88.
Random sorting is the benchmark — rarely reality. People choose where to live, and on traits that move δ.

Pick a target parameter

- Three weights. Three signs. Three answers.
- The choice is about *which causal effect you actually want*.
- **Person-weighted** for aggregate human welfare: how many people are affected, by how much?
- **Unit-weighted** when locations have autonomy: each policy unit chooses independently — the unit *is* the agent.

Today's target: the ATT.

The average effect for the units that actually received the treatment.

Lesson

9. Event Studies: PT Made Falsifiable

Event studies are PT falsifications. What is a falsification?

- Popperian reasoning: you don't confirm a theory — you try to falsify it.
- A rival theory earns scrutiny by making a *prediction you can check*.
- Claim a confounder. Derive what *else* it would do. Test that.
- If the prediction holds $\rightarrow$ you haven't confirmed the rival, but you haven't ruled it out either.
- If the prediction fails $\rightarrow$ the rival weakens.

Popperian asymmetry. Failing to reject the rival does not confirm you. But rejecting the rival's prediction is real evidence.

Two forms of falsification

- **Same outcome, different group.** A group your treatment shouldn't reach, but the rival would.
  Min wage on low-wage employment? Re-run on higher-wage workers.
- **Same group, different outcome.** An outcome your treatment shouldn't move, but the rival would.
  Cigarette tax on smoking-related hospitalizations? Re-run on appendicitis and fractures.

PT falsifications come in both forms. Event studies are one (same group, different "outcome" — a period when no treatment happened).

Famous falsification 1: obesity contagion

- Christakis & Fowler (2007): **obesity is contagious** in social networks.
- Cohen-Cole & Fletcher (2008): re-run the same model on **height, acne, headaches** — same Framingham network, same identification.
- All three also show "contagion."

> *We have not been able to become taller by having tall friends.*

— Cohen-Cole & Fletcher (2008)

Same group, different outcome. The result isn't necessarily wrong — the empirical model is.

Famous falsification 2: milk addiction

- Becker–Murphy (1988): serial correlation in cigarette / alcohol $\Rightarrow$ **"rational addiction."**
- Auld & Grootendorst (2004, *J. Health Econ.*): apply the same estimator to **milk, eggs, oranges**.
- All three look "rationally addictive."

Same group, different outcome. Nobody is rationally addicted to milk. ***The identification — not the goods — is what's broken.***

Point: PT makes predictions beyond the event study. Placebo groups and placebo outcomes are the other two forms.

Parallel trends is about $Y(0)$, not "trends" abstractly

The parallel-trends assumption is a statement about an unobserved object:

$$\mathbb{E}\big[Y(0)_t - Y(0)_{t-1} \,\big|\, D{=}1\big] \;=\; \mathbb{E}\big[Y(0)_t - Y(0)_{t-1} \,\big|\, D{=}0\big]$$

- PT is the counterfactual untreated outcome's trend. **Not** a statement about OBSERVED Y trends in general.
- Observed Y for treated post-treatment is $Y(1)$, not $Y(0)$. The PT assumption can't be read off it.
- The confusion comes from saying "trends are parallel." What we mean is "**untreated potential outcome trends are parallel.**"

Why this matters

We never see treated $Y(0)$ post-treatment. Pre-treatment data is the only window onto whether a similar assumption *could* hold post. That's where event studies come in.

Event studies: pre-treatment placebos for PT

The falsification logic. Suppose some confounder both shifts $Y(0)$ trends *and* is associated with treatment timing. That's a PT violation.

- If a similar confounder structure existed *before* treatment, we should see treated and control Y trends diverge in the pre-period too.
- Estimate placebo "effects" at $t = -2, -3, \ldots, -q$ where there is no treatment. If those coefficients are large or trending, we've rejected the assumption that drives the post-period.

Popperian asymmetry

Flat pre-trends *fail to reject* PT — they do not prove it.
Sloped pre-trends *reject* the design.

Pre-treatment placebos are a falsification, not a proof.

An event study is the 2×2 viewed in motion

- Instead of one pre-period and one post-period, use *many*.
- Estimate a separate effect at each event-time relative to treatment.
- Plot them: $\hat{\beta}_{-q}, \ldots, \hat{\beta}_{-2}, \hat{\beta}_{-1}, \hat{\beta}_{0}, \hat{\beta}_{+1}, \ldots, \hat{\beta}_{+q}$.
- The graph tells two stories: **pre-trends** (left of zero) and **dynamics** (right of zero).

Same arithmetic as the 2×2, repeated against a common reference period.

Every coefficient is a 2×2 against $t = -1$

For each event-time k (with $t = -1$ as the reference):

$$\hat{\beta}_k \;=\; \left(\overline{Y}^T_k - \overline{Y}^T_{-1}\right) \;-\; \left(\overline{Y}^C_k - \overline{Y}^C_{-1}\right)$$

- Two long differences, one subtraction — the 2×2 again, with $t = -1$ as the pre-period.
- Pre-period $\hat{\beta}_k$ ($k < -1$): asks whether trends were already diverging.
- Post-period $\hat{\beta}_k$ ($k \geq 0$): the dynamic ATTs.

Drop $k = -1$ as the reference. Multicollinearity won't let you keep all the dummies.

The saturated OLS specification

$$Y_{it} \;=\; \alpha_i \;+\; \tau_t \;+\; \sum_{t \neq t_0 - 1} \beta_t \cdot \big(D_i \cdot \mathbf{1}\{\text{year} = t\}\big) \;+\; \varepsilon_{it}$$

- α_i — unit fixed effects (absorb levels).
- τ_t — calendar-year fixed effects (absorb common shocks).
- One interaction per calendar year, omitting $t_0 - 1$ (the year just before treatment) as the reference.
- Each β_t is the 2×2 of the previous slide: (treated change from $t_0{-}1$ to t) $-$ (control change from $t_0{-}1$ to t).

In a 2×T design the calendar year is the event-time. We save k for staggered treatment, where cohorts have different treatment dates.

A tiny panel where you can do the math in your head

Cell means $\overline{Y}_{g,t}$. Two treated units (T), two controls (C). Treatment turns on in 2000.

Year t	1997	1998	1999	2000	2001
Treated $\overline{Y}_t^T$	7	8	9	13	16
Control $\overline{Y}_t^C$	2	3	4	5	6
$\overline{Y}_t^T - \overline{Y}_t^C$	5	5	5	8	10

- Pre-period: T–C gap is constant at 5. **Parallel trends.**
- True ATT$(2000) = 3$, True ATT$(2001) = 5$.

Each $\hat{\beta}_t$ by hand — same numbers via OLS

Formula: $\hat{\beta}_t = (\overline{Y}^T_t - \overline{Y}^T_{1999}) - (\overline{Y}^C_t - \overline{Y}^C_{1999})$

By hand

t	Calculation	$\hat{\beta}_t$
1997	$(7-9) - (2-4)$	**0**
1998	$(8-9) - (3-4)$	**0**
2000	$(13-9) - (5-4)$	**3**
2001	$(16-9) - (6-4)$	**5**

Via OLS in R

```
library(fixest)
feols(Y ~ i(year, treat,
            ref = 1999) | unit + year,
      data = toy)
```

coefficient	estimate
`year::1997:treat`	0
`year::1998:treat`	0
`year::2000:treat`	3
`year::2001:treat`	5

OLS matches the manual 2×2s to machine epsilon. The regression is the formula.

LESSON

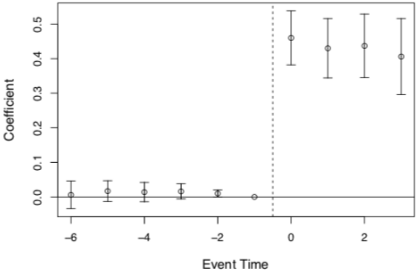
10. Five Pieces of a Strong DiD

Five pieces of a strong DiD

1. **Bite** — the policy actually changed what it was supposed to change.
2. **Event study** — pre/post coefficients around treatment.
3. **Falsification** — run a rival theory on a placebo group or outcome and try to reject *it*.
4. **Main result** — the DiD on the actual outcome.
5. **Mechanism** — the causal pathway from D to Y.

The point estimate alone is a case so weak that readers and the public are justified to reject it.

Bite #1: Medicaid take-up jumped in expansion states

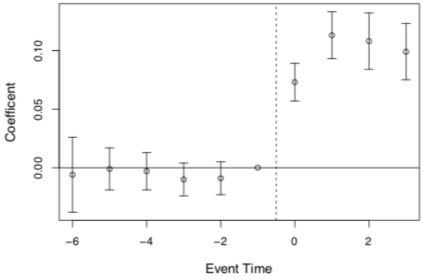


(a) Medicaid Eligibility

Did the policy actually move people onto Medicaid? *Yes — enrollment rose sharply in expansion states post-2014, flat in non-expansion.*

Without bite, there's no story. The mandate has to actually do something before we can attribute downstream changes to it.

Bite #2: any-insurance coverage rose

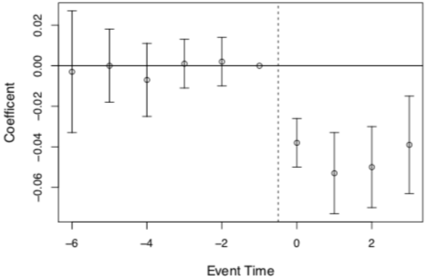


(b) Medicaid Coverage

Did the new Medicaid enrollees actually gain coverage they didn't have? *Or did they just shift from private to Medicaid?*

Any-insurance coverage rose — the expansion brought in the previously uninsured, not the already-insured.

Bite #3: the uninsured rate fell

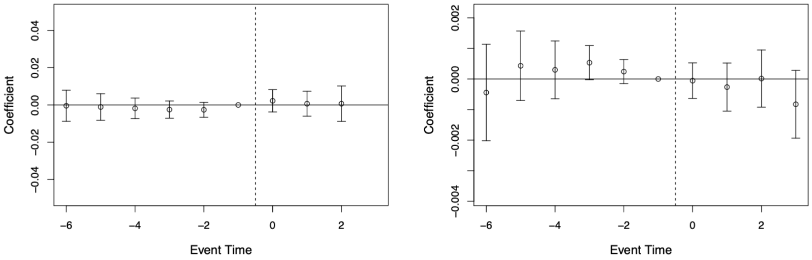


(c) Uninsured

The mirror image of Bite #2. *Uninsured rate dropped sharply in expansion states post-2014. Three measures (Medicaid up, any-insurance up, uninsured down) converge on the same conclusion:* ***the policy bit.***

Falsification: same model, on the 65+ population — no effect

Age 65+ in 2014



(c) Medicaid Coverage (d) Annual Mortality

Same states, same years, same model — but run on the 65+ population. *They all have Medicare, so Medicaid expansion shouldn't touch their mortality.*

A rival theory ("some 2014 shock hit expansion states") would predict a mortality effect here. It didn't appear.

Rival rejected.

Medicaid expansion: $\approx 9.2\%$ reduction in near-elderly mortality

Miller, Johnson & Wherry (2021), *Quarterly Journal of Economics*.

- Did the ACA Medicaid expansion (2014) reduce mortality among the **near-elderly** (55–64)?
- Comparison: states that expanded vs. states that didn't, before vs. after 2014.
- Pre-trends flat. Post-2014 mortality declines in expansion states.
- Effect grows over time — consistent with cumulative treatment for chronic conditions.

Why *near*-elderly?

Everyone 65+ has Medicare — Medicaid expansion is irrelevant for them. The elderly become a natural **placebo group** living in the same economies.

The five pieces, all five present in Medicaid

Bite	Three figures: Medicaid take-up up, any-insurance up, uninsured down
Event study	Pre-trends flat; post-trends decline in expansion states
Falsification	No effect on elderly (65+) — same states, same years, same model
Main result	$\approx 9.2\%$ reduction in mortality among the near-elderly
Mechanism	Disease-related deaths fall and compound; consistent with treated chronic conditions

None of these alone is the case. ***Together they are powerful.***

What we've done today — and tomorrow

Today:

1. DiD's empirical roots — Semmelweis, Ashenfelter.
2. DiD $=$ calculation $+$ assumption: the 2×2 and parallel trends.
3. Five regressions, one number.
4. Decomposing the 2×2 — the +0 trick and three failure cases.
5. Card and Krueger: parallel trends visualized.
6. Standard errors — vanilla, robust, cluster.
7. Population weighting: pick a target parameter.
8. Event studies as falsifications, by hand and via OLS.

Tomorrow: when the 2×2 stops working — covariates, staggered adoption (Goodman-Bacon, Callaway–Sant'Anna, Sun–Abraham, BJS, dCdH).